A 2.5D Approach to Training Robust 3D Pose Estimators in Specialized Exercise Settings

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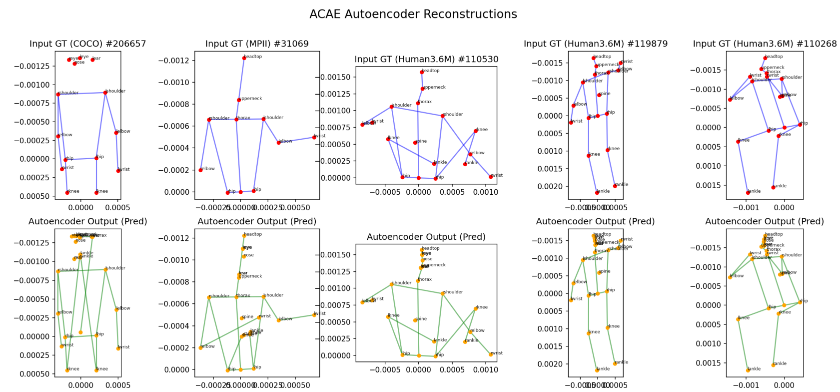

1 Introduction

One of the biggest obstacles in developing robust, generalizable, 3D human pose estimators is the quality of annotated data. Our project seeks to investigate the relationship that diverse 2D supervised data plays when paired with scarce yet valuable 3D labeled data in facilitating more robust 3D pose estimations. Literature has shown promising results in a hybrid 2.5D approach Pavllo et al. (2019); Martinez et al. (2017) that can yield projections that are both anatomically plausible and highly generalizable.

We will evaluate the robustness of this hybrid approach by fine-tuning a pre-trained Spatio-Temporal Transformer Yang et al. (2024) on specialized fitness data with Low-Rank Adaptation (LoRA), custom biomechanical loss functions, and a skeleton-bridging autoencoder that will enable us to use multiple specialized datasets Sárándi et al. (2023). We hope to better understand how "in-the-wild" 2D data can complement "lab-accurate" 3D data to engineer pose estimators that perform reliably in unconstrained, high-stakes environments.

2 Related Work

3D human pose estimation in video with temporal convolutions and semi-supervised training
Pavllo et al. (2019)

- This paper presents a novel approach to reconstructing 3D human pose estimations by "lifting" 2D pose estimations into 3D through a fully convolutional model architecture. This proposed architecture performs dilated temporal convolutions on the predicted 2D pose estimates to output 3D estimates. It also introduces a semi-supervised training methodology via back-projection that is used to circumvent the scarcity of 3D labeled data. This training method projects the predicted 3D pose back into 2D accurately given camera parameters and compares these back-projected points to 2D ground truth, introducing another penalty.

CameraPose: Weakly-Supervised Monocular 3D Human Pose Estimation by Leveraging In-the-wild 2D Annotations
Yang et al. (2023)

- This paper introduces a generator/discriminator pipeline to augment 2D datasets to improve 3D pose estimation. It doesn't require pose labels, and rather uses reprojection back to 2D to minimize the loss there. It introduces a camera parameter branch to the network, such that when 3D poses are estimated, the camera pose is also calculated, and a random transformation can be applied to the data, creating additional 2D data for the model. We can use this to better proportionally weight rare environments by augmenting the occurrence of those environments.

Learning 3D Human Pose Estimation from Dozens of Datasets using a Geometry-Aware Autoencoder to Bridge Between Skeleton Formats
Sárándi et al. (2023)

- This paper poses an autoencoder model that can take multiple 3D datasets and embed them in a latent space such that discrepancies in 3D dataset annotations are reconciled — e.g. we can mix multiple sources to increase the size and diversity of our dataset. The authors report that training on a combination of 3D data led to improved performance on 3D benchmarks, especially the 3D poses in the wild benchmark, compared to prior models trained on one type of dataset. We incorporate components of this framework to allow us to use multiple datasets from specialized domains.

A simple yet effective baseline for 3d human pose estimation
Martinez et al. (2017)

- This paper reports extremely competitive results using a simple architecture that can infer 3D pose coordinates from only 2D joint coordinates. The authors transform 2D imagery data into joint coordinates by effectively treating it as a 2D human pose estimation task, a fundamentally easier task to solve, and the output (and input of the 3D neural network) dataset that is both richer and lower-dimensional.

APTPose: Anatomy-aware Pre-Training for 3D Human Pose Estimation
Yang et al. (2024)

- This paper discusses APTPose, which is a framework that helps improve 3D pose estimation through Hierarchical Masked Pose Modeling and a reprojection module for integrating 2D and 3D supervision. The paper aligns with our proposed method of incorporating hybrid supervision to bridge the domain gap between lab-based datasets and the in-the-wild datasets. A notable difference, though, is that our approach uses LoRA to efficiently specialize a spatio-temporal transformer for specific occlusion patterns of gym exercises.

Our method differs from prior works like the Martinez et al. baseline by incorporating temporal context, beats VideoPose3D's convolutional approach with flexible self-attention mechanisms instead, and improves upon pure CameraPose or skeleton-bridging autoencoders by focusing on robustness against physical constraints rather than just general-purpose accuracy.

3 Problem Decomposition & Detailed Technical Approach

High-Level Strategy: Hybrid Spatio-Temporal Lifting
The proposed strategy utilizes a two-stage transfer learning and domain adaptation pipeline designed to bridge the gap between laboratory-accurate 3D labels and diverse, in-the-wild sequences. We derive this approach and portions of architecture from APTPose described in Yang et al. (2024).

The core of the method is a 2.5D hybrid training procedure that optimizes the model using a mix of fully supervised 3D data (Human3.6M) and semi-supervised 2D gym data. We plan to prevent the model from "forgetting" general human motion by implementing Low-rank adaptation (LoRA) on the attention layers. The 2D keypoints will be pre-extracted using a detector to serve as pseudo-labels for a back-projection consistency check. We discuss specific datasets in the next section.

For this milestone, we focused on establishing a working baseline pipeline and testing with hybrid supervision ideas before implementing the full proposed architecture above. To start, we used the VideoPose3D codebase Pavllo et al. (2019) to train and evaluate a baseline model on Human3.6M. We then built an end-to-end 2D gym video pipeline through extracting detector generated keypoints from fitness videos, converting into required input format, and integrating them into the training loop as an unlabeled 2D branch. This allowed us to run an initial baseline vs. baseline + 2D comparison and assess the feasibility of our proposed method.

This entire approach will be wrapped in an autoencoder network (trained independently for this milestone), which our design necessitates in order to bridge disagreeing skeleton coordinate systems arising from using multiple 2D datasets labeled by different authors Sárándi et al. (2023). During training, this network learns a single shared high-dimensional latent representation of the skeletons. After the core model estimates pose in this latent space, the poses are returned to each skeleton's original coordinate system to compute the reprojection loss.

Neural Network Specifications

The core model takes a video or temporal sequence of 2D joint coordinates and outputs a sequence of 3D coordinates in a root-relative camera coordinate system. The model is optimized using a composite objective function ($\mathcal{L}$) that balances accuracy, generalizability, and physical plausibility:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{3D} + \lambda_1 \mathcal{L}_{proj} + \lambda_1 \mathcal{L}_{biomech} \tag{1}$$

1. $\mathcal{L}_{3D}$ **Supervised 3D Loss:** Mean per-joint position error (MPJPE) between predicted 3D poses and ground-truth labels from laboratory datasets.
2. $\mathcal{L}_{proj}$ **Weakly-Supervised Reprojection Loss:** Applied to unlabeled gym videos. This measures the distance between the input 2D keypoints and the 3D predictions after they are projected back into 2D space via a perspective camera model.
3. $\mathcal{L}_{biomech}$ **Biomechanical Constraint Loss:** A combination of bilateral symmetry loss to enforce equal bone lengths between left and right limbs and Anatomical Hinge Loss to penalize joint rotations that exceed physical human limits, such as knee hyperextension.

Evaluation Metrics

1. **MPJPE / P-MPJPE / N-MPJPE / MPJVE:** MPJPE measures average joint position error, P-MPJPE measures error after Procrustes alignment, N-MPJPE measures error after scale normalization, MPJVE measures velocity error when assessing temporal consistency, and all are measured in millimeters.
2. **Weakly-Supervised Reprojection Loss:** Applied to unlabeled gym videos. This measures the distance between the input 2D keypoints and the 3D predictions after they are projected back into 2D space via a perspective camera model.
3. **Bilateral Length Inconsistency (BLI):** A custom metric measuring the variance in bone lengths to quantify skeletal realism and symmetry.

Hypotheses

- By introducing 2D gym data into the training loop, we hypothesize that the model will expand its "pose understanding" to include more extreme poses that are absent in standard 3D datasets. The 3D data provides the geometric scale understanding, while the 2D gym data provides the necessary range of possible cases.
- We hypothesize that by including biomechanical losses, we prevent the model from "cheating" during 2D reprojection and force the model to produce a skeleton that is physically accurate.
- While incorporating the skeleton-bridging autoencoder means we would sacrifice label accuracy for more specialized training data, we hypothesize that this will be a net performance gain against our general baseline pretrained on generalized 3D datasets.

4 Experiments & Results

Dataset implementation progress
For the current pipeline, Human3.6M is used as the main supervised 3D dataset and serves as the benchmark for our baseline and milestone comparisons. We have also obtained a gym video dataset from Kaggle, which provides additional unlabeled 2D supervision through keypoints generated with detectron2. We are also planning on incorporating the COCO dataset for more 2D supervision in the future. Fit3D is being set up as an evaluation dataset targeting the fitness domain. It contains exercise videos and their 3D joints ground truths provided in a lab environment. We have verified the Fit3D file structure, loaded video and ground truth joint files, and ran inference on sample sequences, but we do not have a Fit3D evaluation metrics yet due to lack of information on the mapping from Fit3D joints to Human3.6M joints. Later in the report we will also mention our skeleton-bridging autoencoder that is relevant to this issue.

Dataset	Description	Size
Human 3.6M	3D labeled, highly controlled laboratory environment	3.6M images
MPII	2D labeled, mostly singular subject images, complex tasks	25K images
COCO - WholeBody	2D, highly variable in-the-wild	250K images
AMASS	3D, diverse motion environment, SMPL format	11,000 motions
Fit3D (Evaluation Only)	3D labeled, gym setting and tasks	3M+ images

Table 1: Datasets Used

Baseline
For our main baseline during this stage of the project, we trained and evaluated VideoPose3D on Human3.6M. It is a standard temporal lifting model with strong performance, and our proposed method can be viewed as an extension of its semi-supervised 2D training potential. With only training Human3.6M for 30 epochs we obtained the following evaluation results with Human3.6M (averaged across categories):

Metric	Error
MPJPE	47.3mm
P-MPJPE	36.9 mm
N-MPJPE	45.3 mm
MPJVE	2.82 mm

Table 2: Baseline Metrics

Above values can serve as a reference point for later experiments. This shows that our base training and evaluation pipeline is functioning correctly.

Baseline + 2D Gym Video
After evaluating with the baseline model, we tested a hybrid supervision experiment by adding a processed gym video 2D branch to the original training setup. While Human3.6M remained as a supervised 3D source, the gym videos served as additional unlabeled 2D keypoint sequences. After training for 10 epochs with the modified training loop on top of the original baseline model, we obtained:

Metric	Error
MPJPE	51.7mm
P-MPJPE	39.0 mm
N-MPJPE	48.8 mm
MPJVE	2.84 mm

Table 3: Baseline + 2D Gym Video Metrics

Compared with the baseline results, the modified model shows observable decline in performance. MPJPE increased by 4.4mm, P-MPJPE increased by 2.1mm, N-MPJPE increased by 3.5 mm, and MPJVE increased by 0.02mm. Although this was not the improvement we initially expected, it is a constructive milestone result demonstrating the trainability of the 2D branch on top of the 3D supervised. It also showed that naively adding noisy fitness domain 2D supervision does not automatically improve the benchmark result.

The addition of the 2D branch did not improve over the 3D only baseline, but still offered useful insight. A possible reason is that the amount of added fitness video data is limited. Another is that the gym videos are not standardized, which includes variations in viewpoint, cropping, noises, etc. Due to mentioned factors, the additional 2D branch may potentially hurt the baseline performance. The current setup supports hyperparameter tuning and longer training, which will be a possible approach to improving model performance. After assessing our datasets, we might also include a split of 2D data from the Fit3D dataset to provide a more standardized 2D video source.

Autoencoder Progress
At this current checkpoint, the affine combining skeleton-bridging autoencoder was trained independently on the mixture of 2D and 3D data (fig. 1), in which loss was calculated as the mean absolute error (MAE) between the input and decoded annotated coordinates after flattening it into 2D space (effectively ignoring missing Z joint coordinates in 2D datasets). The model achieved an MAE of 7.03e-07 after convergence.

5 Next Steps

The primary architectural goal for the next phase is migrating from the VideoPose3D baseline to the DSTformer backbone based on a spatio-temporal transformer model such as in motionBERT. We will apply LoRA fine-tuning to the attention layers at rank 8, allowing the model to specialize on fitness domain occlusion patterns while preserving general anatomical priors. Alongside this, we will implement the full composite loss by adding the reprojection branch for gym videos via our PerspectiveCamera model and the biomechanical regularizer combining bilateral symmetry and anatomical hinge constraints. We expect the biomechanical term to be especially important given the performance regression observed when naively adding the 2D branch, as it should prevent the model from collapsing depth to satisfy the reprojection loss cheaply.

Integrating the skeleton-bridging autoencoder end-to-end with the DSTformer requires addressing two issues identified during independent training. Because the autoencoder was trained to reconstruct only 2D coordinates, it never received a learning signal for depth, leaving its Z-dimension estimates unconstrained. To correct this, we will jointly train the autoencoder with the DSTformer using Human3.6M's 3D ground truth labels, so the model is explicitly penalized for poor depth estimates. We will also apply inverse-frequency batch sampling to balance training across datasets, preventing Human3.6M, which is far larger in size than MPII or the gym video set, from dominating the training and causing the shared latent representation to overfit to lab-captured motion.

To improve the quality of the 2D gym supervision branch, we will augment the gym video pool with COCO keypoint sequences, or possibly 2D aspect of Fit3D, for more standardized in-the-wild coverage, and implement a confidence-weighted reprojection loss that down-weights occluded or unreliable pseudo-labels by detector confidence score. We will also finalize the fit3d25_to_h36m17() joint mapping in prepare_fit3d.py to establish a complete Fit3D evaluation pipeline, which will replace Human3.6M as our primary out-of-distribution benchmark for the final report.

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